

Reinforcement Learning

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1 What is Reinforcement Learning?

Reinforcement Learning is an area of machine learning that seeks to train models to take actions in a dynamic, stochastic environment in order to maximize a cumulative expected reward.

2 Key Concepts

Definition 2.1. A **Markov Decision Process** is a 4-tuple (S, A, P, R) , where:

- S is a set of states
- A is a set of actions
- P is a probability measure such that $P(s_{t+1} = s' | s_t = s, a_t = a)$ is the probability of transitioning to state s' at time $t + 1$ given that the state at time t is s and the agent has performed action a at time t .
- R is a reward function. The agent receives the immediate reward $R(s, a)$ for performing action a in state s .

Definition 2.2. A decision policy is a function $\pi : S \times A \rightarrow [0, 1]$ such that

$$\int_A \pi(s, a) da = 1 \quad \forall s \in S \quad (1)$$

The space of all decision functions is denoted Π .

Given a discount rate $\gamma \in (0, 1)$ and an initial state s_0 , the objective of the decision agent is:

$$\max_{\pi \in \Pi} \mathbb{E}_{s_{1:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | \pi, s_0 = s \right] \quad (2)$$

Definition 2.3. The state-value of a policy π for a Markov Decision Process (S, A, P, R) is a function $V^\pi : S \rightarrow \mathbb{R}$ defined as

$$V^\pi(s) = \mathbb{E}_{s_{1:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | \pi, s_0 = s \right] \quad (3)$$

Note that

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{s_{1:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | \pi, s_0 = s \right] \\ &= \int_A R(s, a) \pi(s, a) da + \mathbb{E}_{s_1, a_0} \left[\mathbb{E}_{s_{2:\infty}, a_{1:\infty}} \left[\sum_{t=1}^{\infty} \gamma^t R(s_t, a_t) | \pi, s_1 = s' \right] | \pi, s_0 = s \right] \\ &= \int_A R(s, a) \pi(s, a) da + \gamma \mathbb{E}_{s_1, a_0} [V^\pi(s_1) | \pi, s_0 = s] \end{aligned} \quad (4)$$

Definition 2.4. The action-value of a policy π for a Markov Decision Process (S, A, P, R) is a function $Q^\pi : S \times A \rightarrow \mathbb{R}$ defined as

$$Q^\pi(s, a) = \mathbb{E}_{s_{1:\infty}, a_{1:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) | \pi, s_0 = s, a_0 = a \right] \quad (5)$$

Definition 2.5. The advantage function of a policy π for a Markov Decision Process (S, A, P, R) is the function $A^\pi : S \times A \rightarrow \mathbb{R}$ defined as

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s) \quad (6)$$

3 Policy Gradient Theorem (PGT)

Definition 3.1. For $s, s' \in S$, let $\rho^\pi(s \rightarrow s', 0) = \mathbb{1}\{s = s'\}$ and $\rho^\pi(s \rightarrow s', 1) = \int_A \pi(a, s) p(s' | s, a) da$. For $s, s' \in S$ and $k \geq 1$, define $\rho^\pi(s \rightarrow s', k+1)$ recursively by

$$\rho^\pi(s \rightarrow s', k+1) = \int_S \rho^\pi(s \rightarrow x, k) \rho^\pi(x \rightarrow s', 1) dx \quad (7)$$

Definition 3.2.

$$J(\theta) = \int_S p_0(s_0) V^\pi(s_0) ds_0 = \int_S p_0(s_0) \int_A \pi(s_0, a_0, \theta) Q^\pi(s_0, a_0) da_0 ds_0 \quad (8)$$

Theorem 1. *Policy Gradient Theorem*

$$\nabla_{\theta} J(\theta) = \int_S \rho^{\pi}(s) \int_A \nabla_{\theta} \pi(s, a, \theta) Q^{\pi}(s, a) da ds \quad (9)$$

where

$$\rho^{\pi}(s) = \int_S \sum_{t=0}^{\infty} \gamma^t p_0(s_0) \rho^{\pi}(s_0 \rightarrow s, t) ds_0 \quad (10)$$

See Appendix for the proof of the PGT.

Note that

$$\rho^{\pi}(s) = \int_S \sum_{t=0}^{\infty} \gamma^t p_0(s_0) \rho^{\pi}(s_0 \rightarrow s, t) ds_0 = \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \pi) \quad (11)$$

so that

$$\begin{aligned} \nabla_{\theta} J(\theta) &= \int_S \rho^{\pi}(s) \int_A \nabla_{\theta} \pi(s, a, \theta) Q^{\pi}(s, a) da ds \\ &= \sum_{t=0}^{\infty} \gamma^t \int_S \int_A P(s_t = s | \pi) \pi(s, a, \theta) \nabla_{\theta} \ln \pi(s, a, \theta) Q^{\pi}(s, a) da ds \\ &= \sum_{t=0}^{\infty} \gamma^t \int_S \int_A P(s_t = s, a_t = a | \pi) \nabla_{\theta} \ln \pi(s, a, \theta) Q^{\pi}(s, a) da ds \\ &= \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \ln \pi(s_t, a_t, \theta) Q^{\pi}(s_t, a_t) | \pi \right] \end{aligned} \quad (12)$$

Additionally, as note in [SML⁺15], for any function $f(s_{0:t}, a_{0:t-1})$,

$$\begin{aligned} &\mathbb{E}_{s_{0:\infty}, a_{0:\infty}} [\nabla_{\theta} \ln \pi(s_t, a_t, \theta) f(s_{0:t}, a_{0:t-1}) | \pi] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [\mathbb{E}_{s_{t+1:\infty}, a_{t:\infty}} [\nabla_{\theta} \ln \pi(s_t, a_t, \theta) f(s_{0:t}, a_{0:t-1}) | \pi, s_{0:t}, a_{0:t-1}] | \pi] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [f(s_{0:t}, a_{0:t-1}) \mathbb{E}_{s_{t+1:\infty}, a_{t:\infty}} [\nabla_{\theta} \ln \pi(s_t, a_t, \theta) | \pi, s_{0:t}, a_{0:t-1}] | \pi] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [f(s_{0:t}, a_{0:t-1}) \mathbb{E}_{a_t} [\nabla_{\theta} \ln \pi(s_t, a_t, \theta) | \pi, s_t] | \pi] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} \left[f(s_{0:t}, a_{0:t-1}) \cdot \int_A \frac{\nabla_{\theta} \pi(s_t, a, \theta)}{\pi(s_t, a, \theta)} \pi(s_t, a, \theta) da | \pi \right] \\ &= \mathbb{E}_{s_{0:t}, a_{0:t-1}} [f(s_{0:t}, a_{0:t-1}) \cdot 0 | \pi] \\ &= 0 \end{aligned} \quad (13)$$

It follows that for any function $f(s_{0:t}, a_{0:t-1})$,

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \ln \pi(s_t, a_t, \theta) [Q^{\pi}(s_t, a_t) - f(s_{0:t}, a_{0:t-1})] | \pi \right] \quad (14)$$

In particular, this implies

$$\begin{aligned}\nabla_{\theta} J(\theta) &= \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \ln \pi(s_t, a_t, \theta) [Q^{\pi}(s_t, a_t) - V^{\pi}(s_t)] | \pi \right] \\ &= \mathbb{E}_{s_{0:\infty}, a_{0:\infty}} \left[\sum_{t=0}^{\infty} \gamma^t \nabla_{\theta} \ln \pi(s_t, a_t, \theta) A^{\pi}(s_t, a_t) | \pi \right]\end{aligned}\tag{15}$$

Analysis in [GBB04] shows that when estimating $\nabla_{\theta} J(\theta)$ from a sample path of the MDP, using the advantage function results in the lowest possible variance of the estimator.

4 Policy Gradient Methods

4.1 Basic Policy Gradient Method

All policy gradient methods use a version of the same two-step procedure:

1. Estimate the gradient of the policy function
2. Use the estimated gradient in a stochastic gradient ascent algorithm

The estimate of the gradient typically takes the form

$$\hat{g} = \hat{\mathbb{E}}_t \left[\nabla_{\theta} \ln \pi_{\theta}(s_t, a_t, \theta) \hat{A}_t \right]\tag{16}$$

4.2 Trust Region Policy Optimization

Trust Region Policy Optimization was introduced by [SLM⁺15]. TRPO optimizes

$$\text{maximize}_{\theta} \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(s_t, a_t, \theta)}{\pi_{\theta_{old}}(s_t, a_t, \theta)} \hat{A}_t \right]\tag{17}$$

4.3 Proximal Policy Optimization

Proximal Policy Optimization was introduced in [SWD⁺17]

5 Q-learning

Q-learning is an approach to reinforcement learning that estimates the action-value function of the optimal policy. Q-learning represents the action-value function as a function, $Q(s, a, \theta)$, typically a deep neural network, with a vector of parameters θ . [MKS⁺15]

Algorithm 1: Q-learning with experience replay

Input: Replay memory D with capacity N
Action-value function Q with random weights θ
Target action-value function $\hat{Q}$ with weights $\theta^- = \theta$

Output: Weights θ

```
1 for  $episode = 0$  to  $M$  do
2   Sample  $s_0$  from emulator
3   for  $t = 0$  to  $T$  do
4      $a_t = \begin{cases} \text{select random action} & \text{with probability } \epsilon \\ \operatorname{argmax}_a Q(s_t, a_t, \theta) & \text{with probability } 1 - \epsilon \end{cases}$ 
5     Execute action  $a_t$  in emulator, observe reward  $r_t$  and state  $s_{t+a}$ 
6     Store  $(s_t, a_t, r_t, s_{t+1})$  in replay memory  $D$ 
7     Sample random minibatch of transitions  $(s_j, a_j, r_j, a_{j+1})$  from  $D$ 
8     Set  $y_j = \begin{cases} r_j & \text{if episode terminates at step } j + 1 \\ r_j + \gamma \max_a \hat{Q}(s_{j+1}, a, \theta^-) & \text{otherwise} \end{cases}$ 
9     Perform a gradient descent step on  $(y_j - Q(s_j, a_j, \theta))^2$  with respect to  $\theta$ 
10    Every C steps reset  $\theta^- \leftarrow \theta$ 
11 return  $\theta$ 
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References

- [GBB04] Evan Greensmith, Peter Bartlett, and Jonathan Baxter. Variance reduction techniques for gradient estimates in reinforcement learning. 2004.
- [MKS⁺15] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King, Dharmashan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, February 2015.
- [SLM⁺15] John Schulman, Sergey Levine, Philipp Moritz, Michael I. Jordan, and Pieter Abbeel. Trust region policy optimization, 2015.
- [SML⁺15] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation, 2015.
- [SWD⁺17] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.

A Proof of the Policy Gradient Theorem

Proof. Let $\phi(s) = \int_A \nabla_\theta \pi(s, a, \theta) Q^\pi(s, a) da$. Then

$$\begin{aligned}
 \nabla_\theta V^\pi(s) &= \int_A [\nabla_\theta \pi(s, a, \theta) Q^\pi(s, a) + \pi(s, a, \theta) \nabla_\theta Q^\pi(s, a)] da \\
 &= \phi(s) + \int_A \pi(s, a, \theta) \nabla_\theta Q^\pi(s, a) da \\
 &= \phi(s) + \int_A \pi(s, a, \theta) \nabla_\theta \left[r(s, a) + \gamma \int_S p(s'|s, a) V^\pi(s') ds' \right] da \quad (18) \\
 &= \phi(s) + \gamma \int_S \int_A \pi(s, a, \theta) p(s'|s, a) da \nabla_\theta V^\pi(s') ds' \\
 &= \int_S \gamma^0 \phi(s') \rho^\pi(s \rightarrow s', 0) ds' + \gamma \int_S \rho^\pi(s \rightarrow s', 1) \nabla_\theta V^\pi(s') ds'
 \end{aligned}$$

Now suppose that for some $k \geq 0$,

$$\nabla_\theta V^\pi(s) = \sum_{i=0}^k \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' + \gamma^{k+1} \int_S \rho^\pi(s \rightarrow s', k+1) \nabla_\theta V^\pi(s') ds' \quad (19)$$

Then

$$\begin{aligned}
\nabla_\theta V^\pi(s) &= \sum_{i=0}^k \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' + \gamma^{k+1} \int_S \rho^\pi(s \rightarrow s', k+1) \nabla_\theta V^\pi(s') ds' \\
&= \sum_{i=0}^k \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' \\
&\quad + \gamma^{k+1} \int_S \rho^\pi(s \rightarrow s', k+1) \int_S \gamma^0 \phi(s'') \rho^\pi(s' \rightarrow s'', 0) ds'' ds' \\
&\quad + \gamma^{k+1} \int_S \rho^\pi(s \rightarrow s', k+1) \gamma \int_S \rho^\pi(s' \rightarrow s'', 1) \nabla_\theta V^\pi(s'') ds'' ds' \\
&= \sum_{i=0}^{k+1} \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' \\
&\quad + \gamma^{k+2} \int_S \int_S \rho^\pi(s \rightarrow s', k+1) \rho^\pi(s' \rightarrow s'', 1) ds' \nabla_\theta V^\pi(s'') ds'' \\
&= \sum_{i=0}^{k+1} \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' + \gamma^{k+2} \int_S \rho^\pi(s \rightarrow s'', k+2) \nabla_\theta V^\pi(s'') ds''
\end{aligned} \tag{20}$$

Hence, by induction, for all $k \geq 0$,

$$\nabla_\theta V^\pi(s) = \sum_{i=0}^k \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' + \gamma^{k+1} \int_S \rho^\pi(s \rightarrow s', k+1) \nabla_\theta V^\pi(s') ds' \tag{21}$$

Taking the limit as $k \rightarrow \infty$, we have

$$\begin{aligned}
\nabla_\theta V^\pi(s) &= \sum_{i=0}^{\infty} \int_S \gamma^i \phi(s') \rho^\pi(s \rightarrow s', i) ds' \\
&= \int_S \sum_{i=0}^{\infty} \gamma^i \rho^\pi(s \rightarrow s', i) \int_A \nabla_\theta \pi(s', a, \theta) Q^\pi(s', a) dad s'
\end{aligned} \tag{22}$$

Plugging this into the definition of $J(\theta)$ yields

$$\begin{aligned}
\nabla_\theta J(\theta) &= \int_S p_0(s_0) \nabla_\theta V^\pi(s_0) ds_0 \\
&= \int_S p_0(s_0) \int_S \sum_{i=0}^{\infty} \gamma^i \rho^\pi(s_0 \rightarrow s, i) \int_A \nabla_\theta \pi(s, a, \theta) Q^\pi(s, a) dad s ds_0 \\
&= \int_S \rho^\pi(s) \int_A \nabla_\theta \pi(s, a, \theta) Q^\pi(s, a) dad s
\end{aligned} \tag{23}$$

□